



Smart Poultry Farming: A Mobile-Based IoT System for Real-Time Broiler Monitoring and Management

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Abstract

The integration of digital technologies in broiler chicken farming offers practical solutions to persistent challenges such as environmental monitoring, feed management, and livestock health. This study presents the development of a mobile-based broiler farming information system integrated with Internet of Things (IoT) devices to enhance operational efficiency, particularly for small- and medium-scale farmers. The system provides real-time monitoring and control via mobile devices, featuring environmental condition tracking, feed and water level notifications, inventory management, and performance analytics. There was a use of waterfall software development methodology with phases of requirements analysis, system design, implementation, testing, deployment, and maintenance. The system makes use of an ESP32 microcontroller, DHT22 temperature and humidity sensors, MQ-135 ammonia gas sensors, and ultrasonic sensors for feed and water level monitoring. The data from the sensors is transferred to a cloud server and is fetched through a user-friendly mobile application. Tests indicate that the system supports effective real-time monitoring of coop conditions, feed and water control, and prompt response to environmental variations by means of automated alerts. It also supports decision-making through access to past data and performance reports. Despite these advantages, there are still limitations pertaining to internet connectivity, sensor stability, and farmer computer literacy. The novelty of this research is its comprehensive, mobile-accessible IoT solution that combines environmental monitoring with the large-scale farm management operations, completing shortcomings of earlier systems with specialized, isolated monitoring.

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INTRODUCTION

Information and communication technology (ICT) innovation has revolutionized several sectors, such as animal husbandry and agriculture [1]. Among all the livestock subsectors, broiler farming is most suitable for digitalization, especially with rising global demand for affordable, high-protein meat in response to increasing population and changing eating habits [2]. The Food and Agriculture Organization notes that new digital technologies have greatly improved livestock productivity, traceability, and sustainability, with Precision Livestock Farming (PLF) being one powerful means of combining sensors, robotics, and data analytics to enhance animal welfare and resource efficiency [3].

Notwithstanding such advancements, small and medium broiler producers, particularly in developing countries, continue to face challenges in adopting digital solutions [4]. The principal hindrances are a lack of adequate technical skills, huge capital outlays, and inadequate infrastructure [5]. Empirical evidence finds the prevalence of persistent operational inefficiencies in broiler rearing, particularly in feed management, environmental management, and animal health monitoring [6]. Feed alone accounts for a maximum of 70% of the overall costs of production, and most cases of mismanagement lead to suboptimal growth rates and increased mortality [7]. Aside from this, the majority of farmers still rely on conventional methods of

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health complication identification, hence leads to delayed interventions and enhanced disease [8].

Recent research has explored the use of Internet of Things (IoT) technology and mobile applications in broiler farming as a way to manage some of the operational issues of small to medium-scale farmers [9]. Other studies have developed automated feeding systems, environmental monitoring, and stock control [6]. For instance, Laksono et al. developed an integration of Android apps and Arduino devices to make the feeding process automatic, and Ong et al. developed a sensor-controlled ventilation system to monitor air quality in the barn [10]. Although these technologies have been shown to improve efficiency with real-time tracking, reduced operational costs, and increased precision, most existing systems are still founded on discrete functions and have no unified platform for comprehensive farm management [11].

IoT-enabled technologies offer substantial benefits for broiler farming by enabling real-time monitoring of key environmental parameters such as temperature, humidity, and ammonia levels [13]. Integrating these with mobile platforms allows farmers to receive alerts, visualize data trends, and make informed decisions remotely [14]. Moreover, the use of AI in behavior classification has shown high accuracy in detecting early signs of disease or stress, further enhancing productivity and reducing mortality rates [15]. However, the fragmentation of current technological solutions limits their broader impact and scalability, particularly for MSME (micro, small, and medium enterprise) farmers.

Although various studies have developed IoT-based solutions for specific aspects of broiler farming, most existing systems are modular and address only one or two components, such as feeding or environmental monitoring [6,9,10,16]. This fragmented approach fails to meet the comprehensive needs of broiler farmers, particularly those managing smaller operations [17]. Hence, what is urgently needed is an end-to-end, mobile-first information system that integrates IoT-based environmental monitoring with features of complete farm management like inventory, health records, medication management, and performance reporting [18]. This gap represents the core novelty of this research.

In response, this study proposes the development of a mobile-based broiler farming information system integrated with IoT devices [19]. Designed for small to medium-sized farmers, the system has monitoring and management

capabilities of farm activities, including cage conditions, feeding times, medication administration, inventory control, and finance reports [20],[21]. Real-time environmental sensor data is linked to the mobile application, supporting timely intervention as well as regular performance tracking [22],[23]. Unlike solutions offered by previous, single-dimensional studies, the system developed here integrates multiple aspects such as environmental monitoring, feed and medication management, and performance reporting under a single mobile-based system. Thus, this integrated solution aims to meet the real needs of small- and medium-scale farmers, filling a gap left by previous studies.

By offering an accessible and comprehensive solution, this research aims to enhance operational efficiency, reduce mortality, and support sustainable broiler farming practices. The integrated system not only bridges the gap between existing technologies but also promotes broader technology adoption among MSME farmers.

METHOD

This research uses the Waterfall development model, which follows a linear process through five stages: requirement analysis, design, implementation, testing, and maintenance [26]. The study began by identifying the needs of MSME (Micro, Small, and Medium Enterprise) broiler farmers through interviews and field observations. The research revealed a lack of integrated systems for managing farm operations and real-time environmental monitoring, particularly for temperature, humidity, and ammonia levels, key factors that significantly impact broiler chicken health.

The system was designed to include a mobile application developed using Flutter and Dart, with Firebase serving as the backend for authentication, real-time data, and cloud storage. For IoT integration, DHT22 sensors (for temperature and humidity) and MQ-135 sensors (for ammonia gas) were connected to ESP32 microcontrollers. These devices transmit data to the mobile application using MQTT for sensor communication and HTTP for database access.

Implementation involved developing a mobile application with features to monitor environmental data, manage feed and medicine inventory, record livestock health, and generate sales reports. The IoT devices were programmed using the Arduino IDE to collect and send real-time data to Firebase, which was then displayed within the application.

System testing was conducted in May 2025 at a broiler farm in Sleman, Yogyakarta. The testing focused on feature functionality, data accuracy, response time, and usability. The results showed high sensor accuracy ($\pm 0.5^{\circ}\text{C}$ for temperature, $\pm 2\%$ for humidity), low latency (under 2 seconds), and strong usability (System Usability Scale score of 82.5). Maintenance included system updates and improvements based on user feedback.

This method ensures that the system is functional, scalable, and aligned with the actual needs of farmers, offering a complete mobile solution integrated with IoT to enhance broiler farm management.

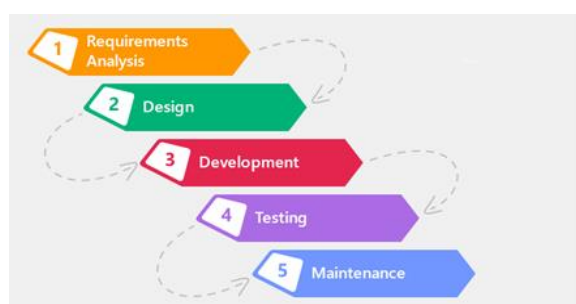


Figure 1. The Stage of Implementing Software

Method of collecting data

This study employed qualitative data collection methods, namely interviews, field observations [27], and document analysis, to gain a comprehensive understanding of the operational conditions and technological needs of MSME (Micro, Small, and Medium Enterprise) broiler farmers [28]. Interviews were conducted with broiler chicken farmers who were actively involved in farm operations, to understand their personal challenges and system requirements [29]. The interviews provided valuable information on prevailing limitations in farm management practices, particularly in regard to the lack of integrated systems for monitoring feed, tracking medication, and controlling the environment. Although no specific number of interviewees is stated, the sample was selected based on relevance to the field of study and a willingness to provide experiential information. In addition to interviews, field observations were carried out at several broiler farms with the objective of witnessing environmental aspects, rearing conditions, and livestock, as well as farmer behavior in person. Observations were particularly directed towards the absence of real-time environmental monitoring systems for such crucial variables as temperature, humidity, and ammonia levels, parameters well

known to have important effects on broiler health and productivity [30]. However, the specific duration of the observation period was not recorded. Document analysis was also utilized to gather supporting data from various written sources, including academic literature, farm records, previous research reports, and, where available, operational documents such as feed logs, health records, and financial statements [31]. These materials served to strengthen the theoretical foundation of the research and support the development of a system tailored to actual field conditions. While the study did not quantify the number of documents analyzed, the information collected played a critical role in validating the field findings and informing system design decisions.

Prototype Design

In general design, researchers create a context diagram to identify the relationship between external entities, inputs, and outputs of this system. With this diagram, we can map the interaction between external entities and the system, without paying attention to the internal details of the system. The interconnected entities are users, IoT devices, cooling fans, and heating systems. Cooling and heating equipment is connected via IoT to the application system, which will later automatically summarize the chicken farming activities added by the user. Users will record all activities in the application system, which will later be able to see a summary of the overall cultivation activities for each harvest.

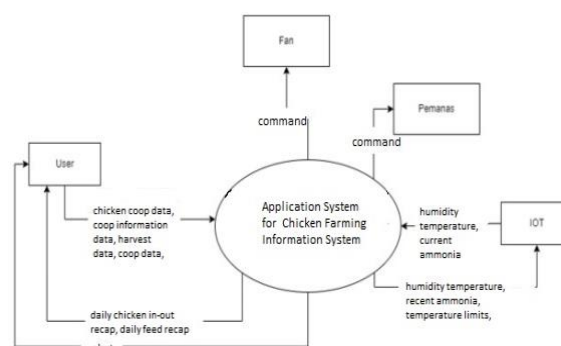


Figure 2. Context Diagram

After the context diagram is well defined (see Fig. 2), the author creates an Entity-Relationship Diagram (ERD) to design a database model to be well visualized between entities of people, objects, and concepts between systems. This ERD produces user entities, IoT, cages, harvests, daily recaps, and feed recaps as depicted in Figure 3.

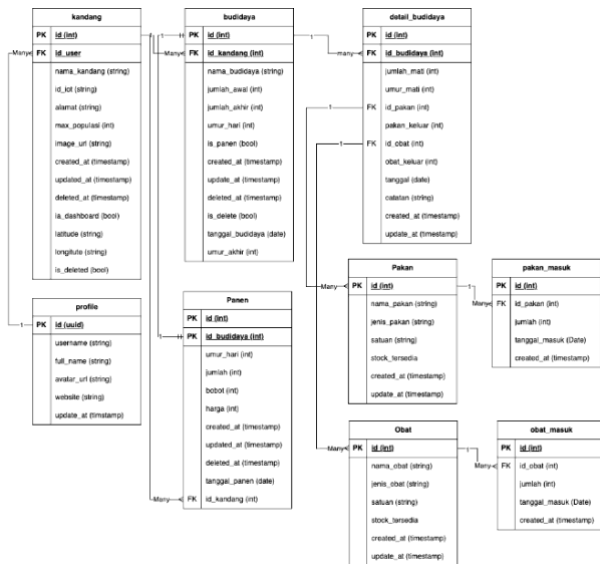


Figure 3. Entity Relationship Diagram (ERD) Diagram

To determine the content and arrangement of data needed to support the system design, the researcher created a database design as shown in Figure 4. To meet the information needs of users, the researcher determined 6 tables to be used, namely the harvest table, the master cage table, the IoT table, the User table, the Feed table, and the Cultivation Information Data Recap table.

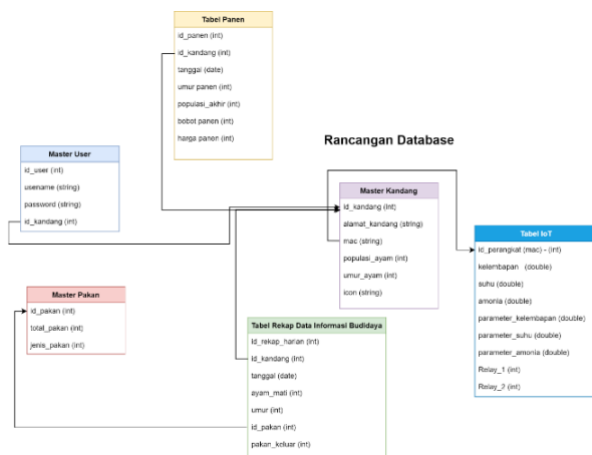


Figure 4. Database Design

To facilitate the system design, the researcher created a use case diagram on the "KandangKu" application as shown in Figure 5, which illustrates the features designed to help farmers manage and monitor their livestock activities. This application has a main actor, namely the farmer, who has full access to various application functions. One of the main features is the Dashboard, which provides a summary report of the condition of the cage and livestock performance, integration of IoT devices for automatic monitoring, and the option to select a

specific cage as the main focus on the dashboard. This feature aims to provide farmers with a comprehensive view of the condition of their livestock.

Next, there is a Cage Management feature that allows farmers to add new cages, view details of existing cages, and update cage information. In addition, there is a Cage Cultivation Management module for each cage, which includes recording livestock cultivation, viewing cultivation details, and recording livestock care and maintenance activities, such as the number of dead livestock, feeding, administering medicine, and other activities that are important for livestock maintenance.

The Feed feature of this app helps farmers to add new feeds, monitor feed stock details, and add stock when they are low, to have feed readily available to livestock. The Medicine feature, however, provides the ability to add new medicines, monitor drug stocks, and add drug stocks to ensure the health of livestock. Besides that, this application also has a Profile feature, which allows users to look at and edit their profile information, and has a feature to shut down the application after use.

Overall, this use case diagram illustrates how the farmers communicate with the other functions within the application, such as cage management, feeds, medicines, cultivation activities, and user profile management. With IoT integration and recording of important activities, this application is expected to help farmers make data-based decisions and improve the efficiency of farm management. This diagram presents a structured map of the application's functional requirements and becomes the main guide in the system development process.

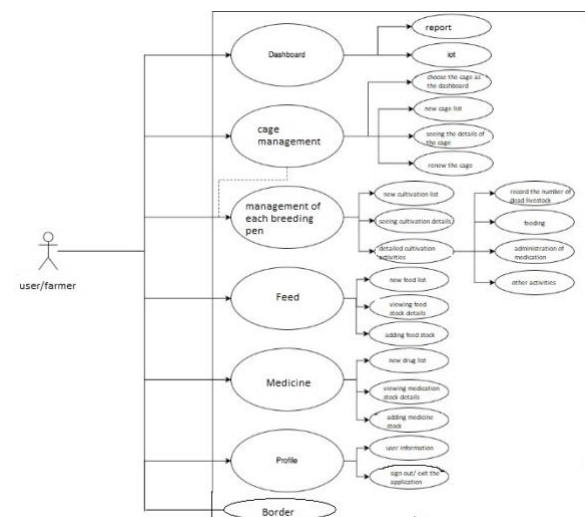


Figure 5. System Development Process

RESULTS AND DISCUSSION

The development of a mobile-based broiler farming information system followed the Waterfall methodology, which allowed for a structured progression through five critical stages: requirements analysis, system design, implementation, testing, and maintenance. This section presents the outcomes of each stage, supported by empirical data gathered from field trials and system performance tests.

Requirements Analysis

The research began with interviews and observations involving broiler chicken producers, particularly MSME farmers, to determine the operational challenges they face. Findings revealed an urgent need for automation and centralized monitoring—particularly regarding feed and water levels, environmental conditions (temperature, humidity, ammonia), and health data—due to the absence of integrated digital systems. These insights guided the system’s functional requirements, especially its mobile-first approach and real-time IoT integration.

System Design

The system was architected to include three core components: (1) a mobile application interface developed using Flutter framework using Dart language, developed via Android Studio or VS Code Studio (2) IoT modules using NodeMCU ESP8266 microcontrollers, DHT11 sensors for temperature and humidity, and ultrasonic sensors for feed/water levels; and (3) a cloud server for historical data storage and synchronization. The application features cage management, feed and medicine tracking, cultivation and harvest modules, and a dashboard for real-time monitoring.

Implementation

Each module was developed in alignment with the design specifications. The IoT components successfully communicated environmental data to the mobile interface via Wi-Fi, while the app enabled real-time control of connected actuators (e.g., fans, lights) and manual override features. Data management, input forms, and responsive UI components were all successfully implemented and integrated.

Testing

The system was tested over 14 consecutive days in a live broiler farming environment. The sensor data demonstrated high accuracy and consistency, with excellent uptime and responsive data transmission. Table 1 presents a sample of environmental sensor readings and system performance metrics:

Table 1. Sensor Readings and System Performance

Date	Temp (°C)	Humidity (%)	Feed Level (cm)	Response Time (ms)	Data Accuracy (%)	Connection Uptime (%)
2025-06-01	32.1	65.3	5.2	820	98.5	99.8
2025-06-02	31.6	66.1	4.8	805	98.7	99.7
2025-06-03	30.8	64.9	4.5	812	98.6	99.9
2025-06-04	31.4	65.5	5.0	798	98.8	99.9

Daily feed consumption was automatically tracked via ultrasonic sensors and presented in the app. The system recorded and calculated feed input, remaining volume, and daily usage with consistent precision (Table 2).

Table 2. Daily Feed Consumption Records

Date	Feed Input (kg)	Feed Remaining (kg)	Consumed (kg)
2025-06-01	50	14	36
2025-06-02	50	12	38
2025-06-03	50	10	40
2025-06-04	50	11	39

Feature-level testing also validated the accuracy and stability of the application’s core functionalities, all of which performed as expected during the trial period (Table 3).

Table 3. Feature Functionality Testing

Feature	Expected Behavior	Actual Outcome	Status
Sensor data updates	Real-time display every 10s	Updates within 10s consistently	Working
Feed/water alert notifications	Alerts when below threshold	Alerts sent accurately at low levels	Working
Data history log	Stores daily values in cloud	Data recorded and exportable	Working
Emergency alerts (temp, humidity)	Notify if values exceed thresholds	Notified via app & SMS	Working
Manual override (actuators)	Turn on fan/feeder manually via app	Manual control successful	Working

Maintenance and User Feedback

Although long-term maintenance data is not yet available, the application's modular design ensures future adaptability. The interface includes full CRUD (create, read, update, delete) functionality for cage, cultivation, feed, medicine, and harvest records. Below is a summarized narrative of key UI modules developed and tested:

- Cage Screen:** Users can add, edit, or delete cages, attach GPS coordinates, and assign IoT devices.
- Dashboard:** Displays environmental metrics, weather data, and device connectivity, and offers quick access to cultivation, harvest, and report features.
- Cultivation Module:** Allows input of initial chicken stock, age tracking, mortality logging, feed and medicine records, and ongoing activity updates.
- Harvest Module:** Enables input and review of harvest data such as weight, final population, and pricing, linked to cultivation records.
- Reports Screen:** Presents chronological records of cultivation and harvest activities with date filters and grouped views.
- User Profile & Auth:** Provides user registration, login, and logout functionality via secure email-password authentication.

Table 4. Summary of App Feature Modules (Excerpt)

Screen	Key Features
Reports	Grouped daily logs, cultivation and harvest history
Authentication	User registration, login, profile management
Cage	Add/edit/delete cages, GPS toggle, IoT device pairing
Dashboard	Real-time IoT display, device control, activity shortcuts
Cultivation	Record initial stock, log mortality, track feed/medicine, view activity log
Harvest	Link to cultivation, record harvest weight/price/final stock

This comprehensive system represents a significant step toward digital transformation. In broiler farming. Unlike existing systems that focus on isolated functions, this study proposes an integrated mobile-IoT platform tailored for small and medium-scale broiler farmers. The seamless coordination of environmental sensing, farm record management, and real-time alerts enables proactive decision-making and data-driven optimization.

**Figure 6.** (a) Screen Cage, (b) Dashboard Screen.

Discussion

Broiler Chicken Farming

In order to achieve maximum yield, broiler rearing is a process that involves many management practices and principles [10]. The fundamentals of broiler rearing consist of several

major issues that span from feeding to healthcare, maintenance practices, and the best possible environmental management. An integral component of broiler rearing is farm management, which includes the techniques for properly managing the daily operations of the farm [33]. This involves weighing and inspecting the hens physically regularly to ascertain that their development is as per the demands of production. Ensuring that there is a tight biosecurity program in place, which includes quarantining new chicks, maintaining cleanliness, and warding off disease-carrying rodents and insects, plays a crucial role in managing chicken health. In addition to this, making sure the right humidity, temperature, ventilation, and lighting levels, which all play roles in chicken growth, requires control of the environment within the home. In order to be able to maintain hens within the most ideal environment under all circumstances, technologies of today, like automation systems and environment sensors, can help monitor and regulate such parameters in real-time [34]. For making the required adjustments toward enhancing productivity and efficiency, correct management also means the collection and analysis of production data to determine trends, issues, and areas for the development of agriculture [36]. Healthy, high-quality broilers to fulfill the requirements of production will therefore be attained when correct farming practices and efficient farm management are applied.

Mobile Application

Developing mobile applications is a dynamic and intricate process that involves choosing the right mobile platform and putting mobile development ideas into practice [37]. In order to make sure that mobile applications are not only useful but also aesthetically pleasing and simple to use, UI/UX (User Interface/User Experience) design is essential. In order to increase user pleasure and engagement, good UI/UX design entails developing responsive interfaces, simple navigation, and pleasurable user experiences. Developers must take into account technologies like iOS and Android when selecting a mobile platform because each has a distinct ecosystem and set of development requirements. The two main platforms for mobile applications are iOS, which utilizes Swift or Objective-C, and Android, which employs programming languages, like Kotlin or Java.

Internet of Things

The Internet of Things (IoT), a theoretical framework that incorporates contemporary technologies to increase productivity and efficiency, is crucial to broiler farming [38]. An internet-connected network of devices that can autonomously gather, send, and share data without the need for direct human contact is known as the Internet of Things (IoT) [39]. This allows various devices, such as sensors and actuators, to operate in an integrated network, so that relevant data can be processed in real-time to support database decision-making. IoT architecture generally consists of several main components: sensors that collect environmental data such as temperature, humidity, and air quality; communication devices that connect sensors to cloud platforms via technologies such as Wi-Fi, Bluetooth, or LoRa; cloud platforms that function as data processing and storage centers; and applications and analytics that enable real-time monitoring, control, and data analysis. In its application in agriculture, especially broiler farming, IoT is used to monitor the cage environment through sensors that measure important parameters, ensuring optimal conditions for chicken growth and health [25]. In addition, IoT also supports animal health management by monitoring the activity and physical condition of chickens, allowing early detection of disease or stress so that appropriate action can be taken. Not only that, IoT facilitates the automation of livestock farming processes, such as automatic feeding and water systems, in ventilation settings, and cooling systems, all of which contribute to reducing manual work, increasing consistency, and operational efficiency [27],[28]. IoT is therefore essential to the development of integrated and technology-based information systems since it makes it possible to manage broiler farming more intelligently, effectively, and responsively to environmental dynamics.

Development of a Mobile-Based Integrated Internet of Things Device for Broiler Chicken Farming Information System

This study shows the way the development of a mobile-telephone-based broiler chicken production information system using Internet of Things (IoT) technology enhances farm management effectiveness, responsiveness, and data-driven decision-making [40]. The system supports real-time monitoring of important environmental conditions such as temperature, humidity, and water and feed levels via sensors connected to a cloud-based system whose

information can be accessed via mobile phones [41].

In contrast to other research that developed IoT-based systems with specific emphasis on some attributes, such as automated feeding [17] and ventilator-aided ventilation [10], the system under discussion integrates a range of functionalities, ranging from environmental monitoring and tracking of feeds and water to inventory management and performance reporting under a single mobile-compatible platform. This integrated system addresses the shortcomings of earlier systems that were typically modular and incapable of fitting holistic farm production [11]. By combining all these attributes, the system presented here is an inexpensive and scalable solution in line with the operating conditions of small and medium-scale broiler farmers.

The system displayed high data accuracy, reliable sensor response, and consistent connectivity under field conditions, thereby enabling automated feeding interventions and real-time alerting of anomalies. The app also supports comprehensive management features, including feed and medicine monitoring, cultivation and harvest documentation, and historical analysis. The integrated features enable the farmer to optimize production, reduce losses, and adopt a more sustainable and technology-driven model of broiler chicken production.

LIMITATIONS

The research should be acknowledged with several limitations, although this research has been successful in developing a mobile-based broiler farming information system application that integrates Internet of Things (IoT) devices. For starters, there still exists limited utilization of the research findings since this application was piloted on a small level and did not reach deployment in other farming settings and geographical areas. Secondly, the integrity of hardware and proper solid internet connectivity are crucial in ensuring the IoT sensors are dependable. The system may not be able to access real-time data collection and transfer in regions of weak network support. Thirdly, the vast majority of small and medium-scale farmers would still lack the technical know-how and information literacy necessary to fully leverage this technology-based application. To enable the target users to embrace and utilize the program successfully, technical support and training are required.

Lastly, despite being of utmost importance in the implementation of IoT-based technologies,

data security and user privacy were not widely researched during system design. The given limitations are for further research to create applications with higher system robustness, a greater number of users, and the incorporation of data security aspects.

CONCLUSION

This study confirms that the development of a mobile-based information system integrated with Internet of Things (IoT) devices significantly enhances the productivity and management of broiler farming. The system provides real-time access to environmental parameters like temperature, humidity, and feed and water status via mobile devices, with the information being transmitted through IoT-connected sensors. Farmers can optimize feed and water provision based on automated data collection and attend to emergencies in good time using system-generated alerts. There is, in addition, a repository of historic data for performance analysis, enabling the early identification of faults in operation and continuous optimization of the productivity of the farm. Unlike existing systems with individual functions, this study proposes an integrated mobile-IoT system for use by small and medium-scale broiler farmers. The system contributes to the overall growth of digital agriculture by facilitating sustainable, data-informed livestock management. However, several implementation challenges remain, for instance, sensor durability, internet connectivity, data privacy, and user digital literacy. These challenges need to be addressed to realize the optimum system effect.

However, this research also has some limitations. First of all, the system was tested over a fairly short time frame (14 days) and in a narrowly specified range of farm environments that might not most accurately reflect its performance in a broad variety of varying seasonal, geographic, or operating conditions. The existing prototype is also primarily focused on environmental monitoring and core management functions, but not predictive analytics or automated decision-making based on AI. Moreover, the existing version lacks multilingual support and robust offline support, both of which are necessary for mass adoption, particularly in rural areas with poor internet connectivity. User feedback was gathered ad hoc and not through a planned usability testing procedure, and therefore, the level of insight into user experience and system usability is limited. Subsequent releases of the system ought to have the capability to rectify these weaknesses in order to make it scalable, robust,

and responsive to the diverse needs of broiler farmers.

Future studies must conduct longer field trials to establish the economic payback and rate of return on investment (ROI) of the system on varying farm sizes and geographic locations. Analysis of using artificial intelligence (AI) for predictive analytics and auto-decision-making would further enhance system functionality. In addition, creating user interfaces that are intuitive and easy to use for farmers and offline functionality in areas where connectivity is poor would increase ease of adoption and use. Finally, future studies should target the solution of cybersecurity and ethical considerations in dealing with data in smart farming environments.

AUTHOR CONTRIBUTIONS

A.K.N. conducted the study design, system development, and drafting of the first version of the article. H.P.W. provided support in data analysis, validation of the system, and editing of the manuscript. Both authors have read and approved the final manuscript.

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